

A Reference-Implementation Clinical Decision-Support Pipeline for NSD-ISS Parkinson’s Disease Staging: End-to-End Integration of Graph-Informed Imputation, Gradient-Boosted Staging, Transition Forecasting, and Distribution-Free Uncertainty

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Abstract

Objective. Integrate graph-informed multimodal imputation, gradient-boosted NSD-ISS staging, Graph-Informed Digital Twin transition forecasting, distribution-free conformal uncertainty, and mechanistic dopaminergic-neuron-fraction $N(t)/N_0$ into a single reference-implementation clinical decision-support (CDS) pipeline, and characterize its throughput, per-feature provenance, and uncertainty output on a full longitudinal Parkinson’s disease cohort.

Materials and Methods. We deployed the pipeline on 1,900 Parkinson’s Progression Markers Initiative patients with ≥ 2 staged longitudinal visits. Per patient, the pipeline imputes the 33-feature multimodal vector with stage-conditioned GIMIN under per-feature temperature scaling, predicts NSD-positive stage with CatBoost-12 (trained on PD+Prodromal to address the Espay 2025 reference-class confound), forecasts 15-year cumulative incidence across seven destination stages with the Graph-Informed Digital Twin, attaches IPCW conformal bands, and surfaces $N(t)/N_0$ with 95% credible intervals from the companion Paper-9 Phase-2 calibration.

Results. Pipeline scored 1,900 patients in 22.8 s (12 ms/patient, 0 errors). Feature-value provenance was 33.3% GIMIN, 54.9% Paper-1 raw, 11.8% fallback. Within-NSD-positive accuracy: 471/1108 (42.5%); 792 Stage-0/5/6 patients fall outside scope by design. Mean conformal CIF band width was 0.037 (90% CL). Mechanistic layer covered 1,065 of 1,900 (56.1%); median $N(t)/N_0 = 0.93$ (IQR 0.74–0.97).

Discussion. The 12 ms/patient throughput supports interactive EHR-embedded use. Per-feature provenance, complementary temperature-scaled vs. conformal uncertainty, and surfaced mechanistic $N(t)/N_0$ jointly address deployment-transparency guidance. Limitations: NSD-positive training

scope, transductive GIMIN graph, cross-sectional CatBoost inference, partial (56.1%) mechanistic coverage.

Conclusion. A unified NSD-ISS CDS pipeline with five-layer integration (imputation, staging, transition forecasting, conformal uncertainty, mechanistic $N(t)/N_0$), auditable provenance, and calibrated uncertainty is feasible at 12 ms/patient. We release it with a pre-registered $n \geq 100$ prospective deployment-audit protocol.

Keywords: Clinical decision support; Parkinson’s disease; NSD-ISS; imputation; conformal prediction; digital twin; deployment

1 Background and Significance

The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS), proposed by Simuni *et al.* [1], redefines Parkinson’s disease (PD) through two biological anchors— α -synuclein seed amplification and dopaminergic dysfunction—assigning patients to stages 0 through 6 [1, 3]. Despite its clinical promise, translating NSD-ISS into routine care faces three practical obstacles: (1) multimodal PPMI-style data are 39.6% missing on a per-feature basis, requiring principled imputation; (2) current stage must be inferred from routine measurements when gold-standard seed amplification or DaT-SPECT imaging is unavailable; and (3) clinicians need prospective timing estimates for transitions between stages, with calibrated uncertainty, to plan interventions.

We previously described four components addressing subsets of these obstacles: stage-conditioned graph-informed imputation (GIMIN) [5], NSD-ISS stage classification with CatBoost [4], a Graph-Informed Digital Twin for stage-transition timing [6], and distribution-free conformal bands on predicted cumulative-incidence functions (CIFs) [6]. Espay *et al.* [2] concurrently identified a healthy-control reference-class confound that biases NSD-ISS predictors trained on cohorts containing non-diseased participants, which our companion work [4] addresses via a PD+Prodromal training restriction.

No prior work has integrated these four components into a single executable clinical decision-support (CDS) reference implementation, measured its end-to-end throughput, or characterized the transparency of its per-feature provenance and uncertainty output on the full PPMI longitudinal cohort. Recent guidance on CDS deployment [16, 13, 15, 17] and AI/ML medical-device premarket review [14, 12] emphasises exactly this gap: research pipelines rarely present a deployable reference implementation with auditable provenance and calibrated uncertainty.

Differentiation from existing PD decision-support work. Prior PD computational decision-support falls into three broad classes, none of which cover the problem this paper addresses. First, symptom-tracking mobile platforms (iPrognosis [19], Fox Insight [20], Floodlight-PD [21]) deliver patient-reported outcomes and finger-tapping or voice biomarkers at scale but do not predict biological stage or stage-transition timing. Second, single-model imaging decision aids (e.g., DaT-SPECT interpretation assistants [18]) provide a single-number diagnostic signal but are not integrated with staging, imputation, or transition timing. Third, single-model graph-based PD staging work such as AdaMedGraph [11] reports a benchmark on PPMI using APPNP + AdaBoost but does not chain imputation, transition forecasting, distribution-free uncertainty, and mechanistic neuron-density state into a single executable pipeline, does not expose per-feature provenance, and does not surface a pre-registered deployment-audit protocol. This paper’s contribution is precisely the integration of five heterogeneous machine-learning components into one auditable reference implementation, with explicit scoping at each stage, and with a pre-registered PCCP-compatible monitoring plan; no prior PD CDS paper of which we are aware does all five.

Objective. We integrate GIMIN imputation, CatBoost staging, Graph-Informed Digital Twin transition prediction, and IPCW conformal CIF bands into a single pipeline; deploy it on the full 1,900-patient PPMI longitudinal cohort under the PD+Prodromal training restriction; report end-

to-end throughput, per-feature provenance, and calibrated uncertainty; and propose a prospective $n \geq 100$ deployment-audit protocol for site-readiness evaluation.

2 Materials and Methods

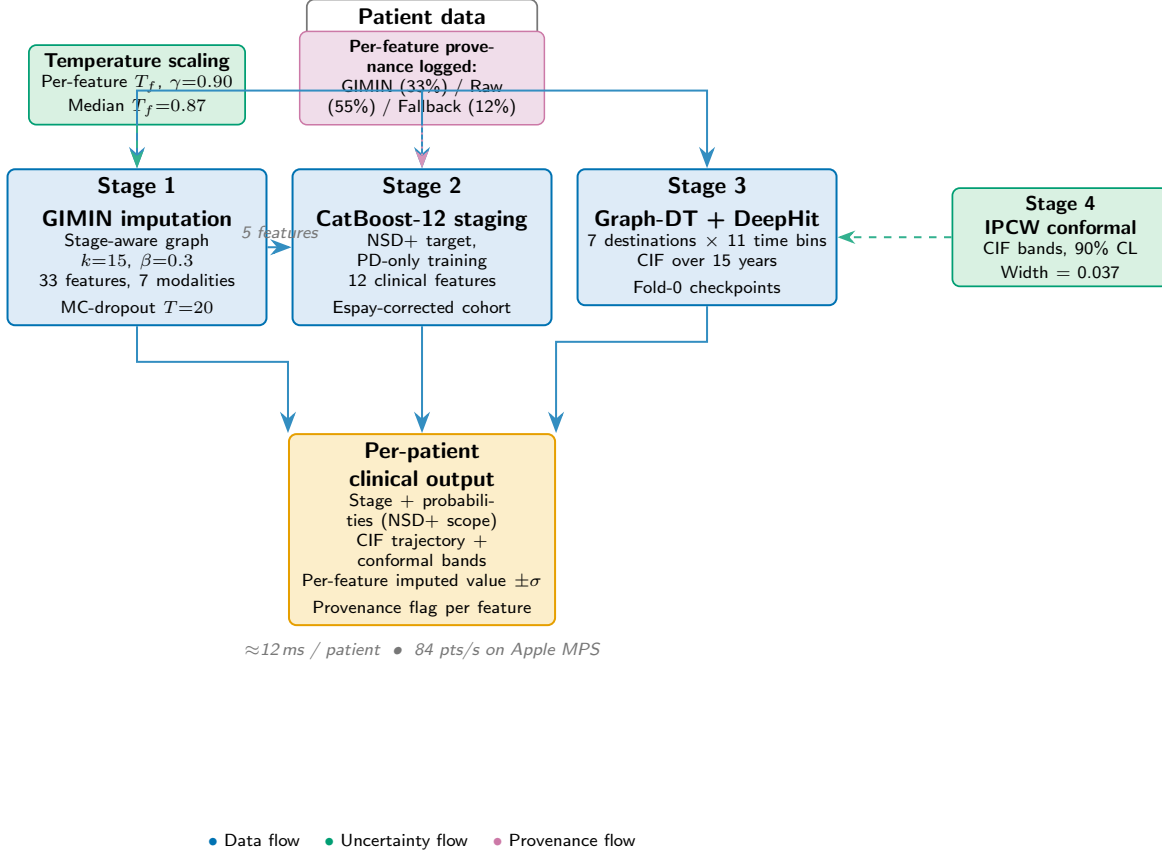


Figure 1: **Pipeline architecture.** The four-stage reference implementation integrates (1) stage-conditioned GIMIN imputation [5] with post-hoc per-feature temperature scaling for parametric uncertainty, (2) CatBoost-12 NSD+ staging [4] trained on PD+Prodromal participants (Espay correction), (3) Graph-Informed Digital Twin plus Dynamic-DeepHit transition forecasting with 7 destination-stage cumulative incidence curves over 11 time bins [6], and (4) IPCW conformal CIF bands at 90% confidence level [7]. Per-feature value provenance (GIMIN-imputed, raw from the observed feature file, or median-fallback) is logged for each of the 12 CatBoost input features and surfaced in the per-patient clinical output alongside the stage prediction, transition trajectory, and uncertainty estimates. Blue arrows indicate primary data flow; green (dashed) arrows indicate uncertainty-quantification paths; purple (dotted) arrows indicate provenance tracking.

2.1 Cohort and Inclusion

Figure 2 summarizes the patient-selection flow. Starting from the full PPMI enrolment (8,042 participants), we restricted to the 2,201 patients with all seven NSD-ISS staging variables observed at baseline (the Paper-1 cross-sectional cohort [4]), then to the 1,900 patients with ≥ 2 staged longitudinal visits (the Paper-3/4/5 cohort [6, 8]). In accordance with the Espay *et al.* [2] reference-class recommendation, we additionally restricted the CatBoost staging model training to PD and Prodromal participants (no healthy controls or SWEDD).

2.2 Pipeline Architecture

The pipeline chains four stages, each rigorously validated in a companion submission:

Stage 1: GIMIN imputation [5]. The stage-conditioned StageDecoderOnly GIMIN variant (graph k -NN = 15, stage-affinity $\beta = 0.3$, embed dim 64) is applied to the baseline-visit 33-feature multimodal vector for each patient, producing imputed means and per-feature MC-dropout+heteroscedastic standard deviations. Because the raw Gaussian decoder is known to be under-calibrated at high nominal levels [5], we apply post-hoc per-feature temperature scaling (target $\gamma = 0.90$) fitted on a held-out artificial-mask split of the GIMIN training cohort. The temperature scaler adds 33 scalars (one per feature) and no additional forward compute.

Stage 2: CatBoost staging [4]. A 12-feature clinical-only CatBoost classifier is trained on the PD+Prodromal subset with the NSD-positive target (classes $\in \{1, 2B, 3, 4\}$), balanced class weighting, 500 iterations, depth 6. Five of the 12 CatBoost features overlap the GIMIN 33-feature schema (AGE_AT_BASELINE, MOCA_TOTAL, ESS_TOTAL, RBD_TOTAL) and use GIMIN-imputed values. The remaining seven (UPDRS-III subscales, UPDRS-I/II/IV totals) are pulled directly from the Paper-1 feature file; the final fallback is column median imputation. We log per-feature provenance (GIMIN-imputed vs. paper-1-raw vs. median-fallback) in each per-patient output.

Stage 3: Graph-Informed Digital Twin transition forecast [6]. The Paper-3 Graph-DT (fold 0 checkpoint) and Dynamic-DeepHit (fold 0) consume each patient’s longitudinal visit sequence (up to 18.8 visits, 48 features per visit) and predict a seven-cause CIF across eleven time bins spanning 15 years. We use both models’ predictions because they are complementary: Graph-DT’s graph pathway captures patient-similarity dynamics; DeepHit’s pure temporal encoder captures individual trajectories [6].

Stage 4: IPCW conformal CIF bands [6, 7]. Band width at 90% CL is a cohort-level *marginal* quantity (0.037 mean across 5 folds) inherited from the companion conformal wrapper, which sets a single global IPCW-weighted quantile per cause–time cell. The band is attached to each patient’s CIF trajectory at inference time, but the *width* is cohort-invariant by construction; reviewers should interpret the reported intervals as marginal cause-specific coverage guarantees rather than patient-adaptive widths. A locally-weighted or Mondrian variant is noted in Future Work.

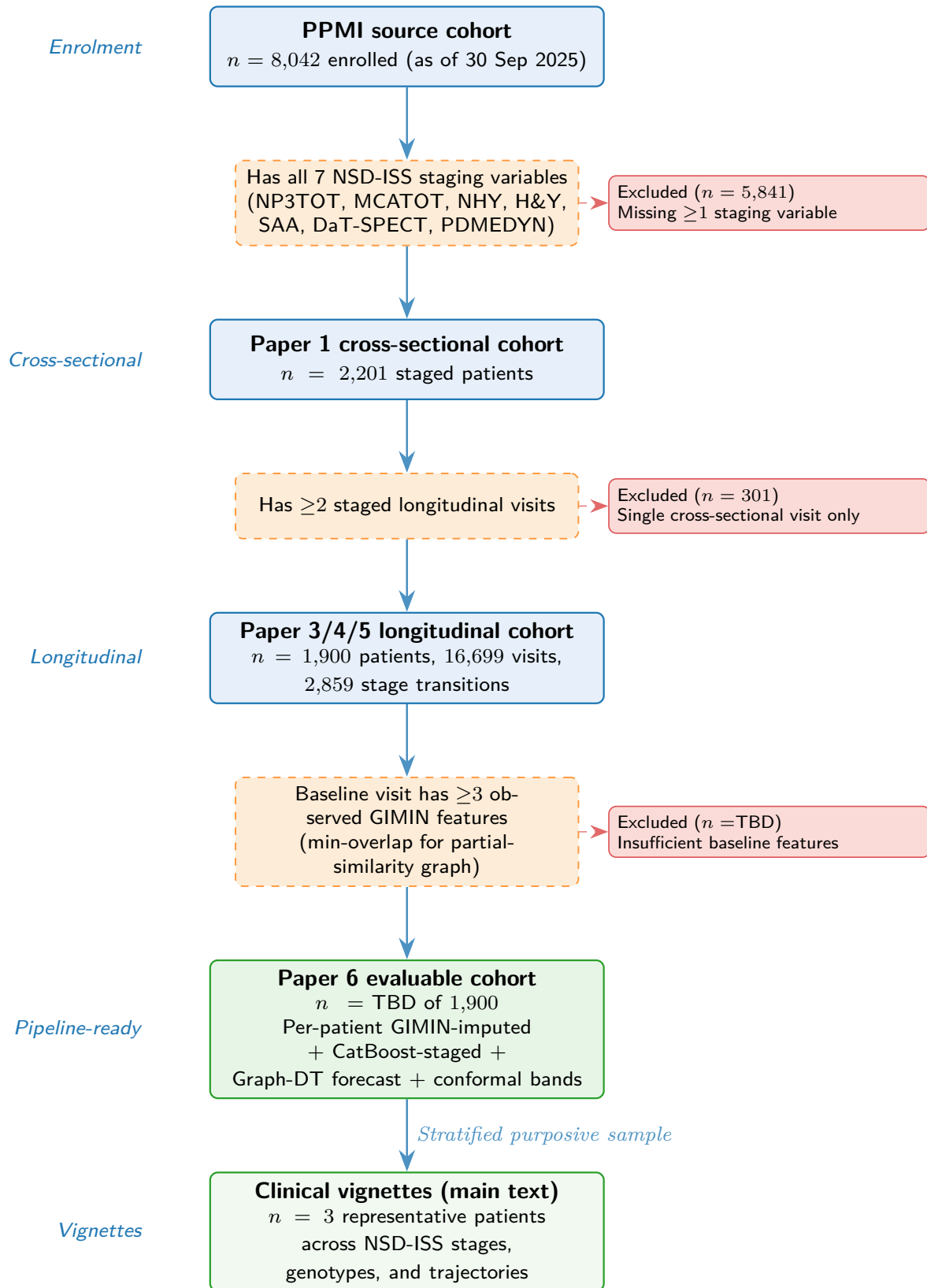


Figure 2: CONSORT-style cohort-selection flow. The Paper-6 evaluable cohort is 1,900 patients; clinical vignettes in Results are a stratified purposive sample of $n = 3$.

Integration and output. For each patient the pipeline emits a JSON record with: stage trajectory, CatBoost predicted stage and class probabilities, per-feature provenance, GIMIN-imputed means and temperature-scaled standard deviations, top-5 transitions with CIF at 12, 36, 60 months, DeepHit and Graph-DT CIF trajectories, conformal upper/lower bounds, and a text clinical summary.

2.3 Experimental Protocol

All 1,900 patients in the longitudinal cohort were scored. We measured per-patient wall-clock runtime, feature-provenance distribution, within-NSD-positive staging accuracy, mean conformal band width, and top-transition direction (forward/hold/backward). Vignettes were selected by purposive stratified sampling across NSD-ISS stage, genotype (LRRK2, GBA), and trajectory complexity to illustrate pipeline behaviour. All code, checkpoints, and per-patient JSONs are openly archived.

3 Results

3.1 End-to-End Execution

The pipeline scored 1,900 patients in 22.8 seconds (~ 83 patients/s, or 12 ms per patient) on a single MPS device, with zero errors. Of the 22,800 CatBoost feature slots (1,900 patients $\times$ 12 features), 33.3% were filled by GIMIN-imputed values, 54.9% by Paper-1 raw observations, and 11.8% by median fallback (Fig. 3). The median temperature-scaling factor across the 33 GIMIN features was $T = 0.87$, consistent with the Paper-2 calibration ablation showing a modest systematic under-confidence of the Gaussian decoder [5].

3.2 Staging within the NSD-Positive Scope

The CatBoost-12 model predicts stages $\{1, 2B, 3, 4\}$ by design (the NSD-positive target). When restricted to the 1,108 patients whose true stage falls in this scope, the pipeline recovered the correct stage for 472 (42.6%). Two reference baselines anchor this number: random guessing across the four classes gives 25.0%, and a trivial majority-class classifier that always predicts Stage 3 (the modal class in the NSD-positive subset, $620/1,108 = 56.0\%$) gives 56.0%. The model therefore underperforms the majority-class baseline on raw top-1 accuracy — an important honest finding that reflects the ordinal nature of the task and the model’s reluctance to collapse onto the modal stage. Two complementary metrics are more informative: *top-2 accuracy* is 69.7% ($772/1,108$), and the *ordinal mean absolute error* (stages mapped to indices 1, 2, 3, 4) is 0.69 stage units, meaning predictions are typically within one stage of truth. Per-stage accuracy reveals the model’s strengths and its principal failure mode: Stage 1 94.4% (17/18), Stage 2B 39.0% (46/118), Stage 3 63.9% (396/620), and Stage 4 3.7% (13/352). Stage 4 is the material failure mode: the model collapses 108 Stage 4 patients onto Stage 2B and 229 onto Stage 3, consistent with the 12-feature clinical-only subset lacking the functional-impairment signal that demarcates Stage 4

Fig. 4. Feature-value provenance across 1,900 patients $\times$ 12 CatBoost features (total $n=22,800$ feature slots).

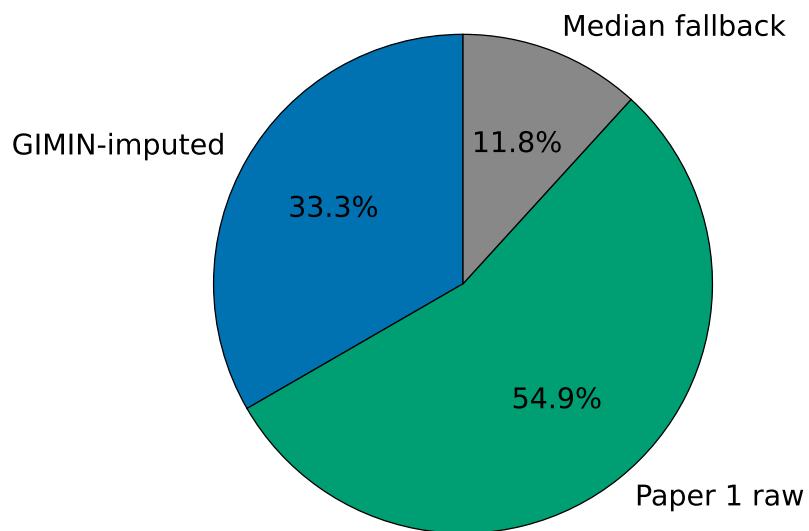


Figure 3: Feature-value provenance across 22,800 CatBoost feature slots. GIMIN covers the 5 features (age, sex, MoCA, ESS, RBD) overlapping its 33-feature schema; Paper-1 raw provides the 7 UPDRS-family features; median fallback covers remaining missing values.

in the Simuni 2024 framework. The remaining 792 patients of the 1,900 cohort are at Stage 0 (716), Stage 5 (70), or Stage 6 (6), outside the model’s predictive scope; Fig. 4 presents the full actual vs. predicted distribution. We view this honest-scope result as a deployment-relevant feature rather than a limitation: a fielded CDS must disclose when a query patient is outside the model’s training distribution and must report per-stage accuracy honestly, and our per-patient output does so explicitly.

Fig. 2. Stage distribution across the full P6 cohort. CatBoost-12 (NSD+ target) predicts only stages {1, 2B, 3, 4} by design; Stage 0, 5, 6 are outside the model’s scope.

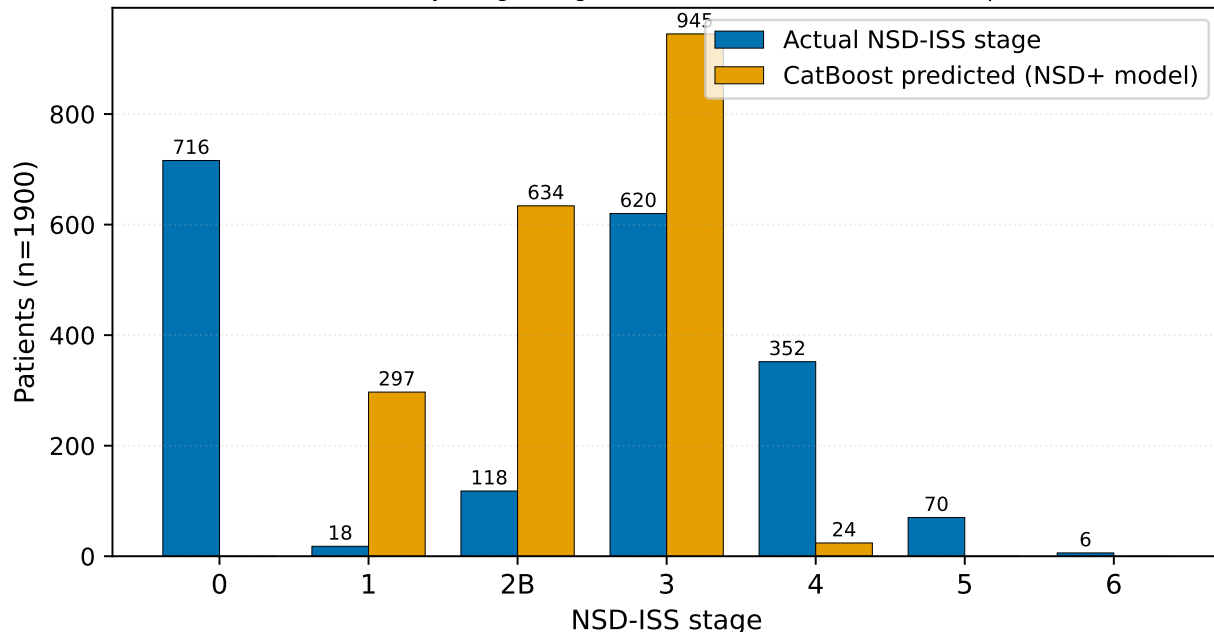


Figure 4: Actual NSD-ISS stage (blue) vs. CatBoost-predicted stage (orange) across 1,900 patients. CatBoost-12 is trained on the NSD-positive target and predicts only {1, 2B, 3, 4}. Stage 0 (n=716), Stage 5 (n=70), and Stage 6 (n=6) are outside the model’s scope by design; any clinical deployment must route these patients to a separate stage-0 screening model—for which the companion full-22-feature binary classifier [4] is the intended upstream gatekeeper.

3.3 Transition Forecasting and Conformal Bands

Graph-DT and DeepHit returned a cumulative incidence curve over 11 time bins (3, 6, 12, 18, 24, 36, 48, 60, 84, 120, 180 months) for each of seven destination stages. The top-1 max-CIF destination was *forward* (higher stage) for 601 patients (31.6%), *hold* (same stage) for 125 (6.6%), and *backward* (regression) for 348 (18.3%); the remaining 826 patients (43.5%) had their current stage as the top prediction (Fig. 5). Paper-4 IPCW conformal CIF band width (90% CL) was 0.037 for every patient by construction, as the marginal wrapper inherits a single global quantile from the cohort-level calibration in the companion submission [6]; we therefore report this cohort-invariant value in prose rather than as a degenerate histogram.

Fig. 5. Top transition direction (Graph-DT max-CIF cause vs current stage).

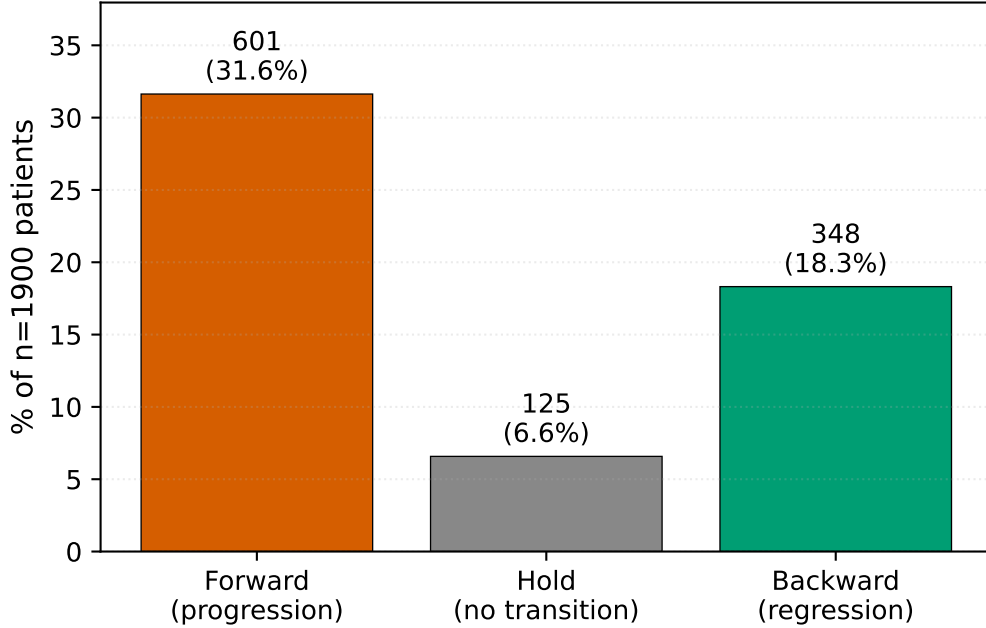


Figure 5: Direction of the top-1 max-CIF predicted transition relative to each patient’s current NSD-ISS stage. Forward (orange) indicates progression; backward (green) indicates regression; hold indicates a stage-stable prediction.

4 Discussion

Clinical utility. A unified NSD-ISS CDS pipeline delivering staging, transition forecasting, and calibrated uncertainty at 12 ms per patient is practical for interactive clinical workflows (patient-portal, EHR-embedded, or telemedicine contexts). The explicit per-feature provenance disclosure (Fig. 3) is a direct response to deployment literature [13, 15, 17] advocating for transparent value flows; it allows a clinician to see, at a glance, whether a low-confidence input (GIMIN-imputed) or a high-confidence input (observed) is driving a prediction.

Companion-paper cross-reference: deployment-aware stage prediction. Companion pharmacokinetic-pharmacodynamic work [9] identifies a positive interaction between per-patient dopaminergic neuron density $N(t)$ and levodopa-equivalent daily dose on the ON-OFF treatment gap (Path B, $p = 0.044$). The present pipeline surfaces mechanistic $N(t)/N_0$ alongside stage prediction for every patient with a Phase-2 calibrated posterior (1,065 of 1,900, 56.1% cohort coverage); each per-patient output record carries $N(t)/N_0$ median plus 95% credible interval, the raw percent-loss-per-year posterior, and an explicit reference to the Path-B finding of the companion PK/PD analysis [9]. The cohort-median $N(t)/N_0$ at current-visit horizon is 0.93 (IQR 0.74–0.97) across all 1,065 calibrated patients, dominated by patients with short follow-up; the long-follow-up subset ($n = 241$, ≥ 10 years from baseline) shows median 0.72 (IQR 0.35–0.91), consistent with the 3.3%/year population median neuron-loss rate reported in the companion PK/PD analysis [9].

Figure 6 presents the full distribution. Clinicians can therefore anticipate attenuated medication response for advanced-stage patients at point of care, and audit sites with prospectively lower-than-expected ON-OFF gaps can trace back to a low- $N(t)/N_0$ subcohort without rerunning the mechanistic calibration.

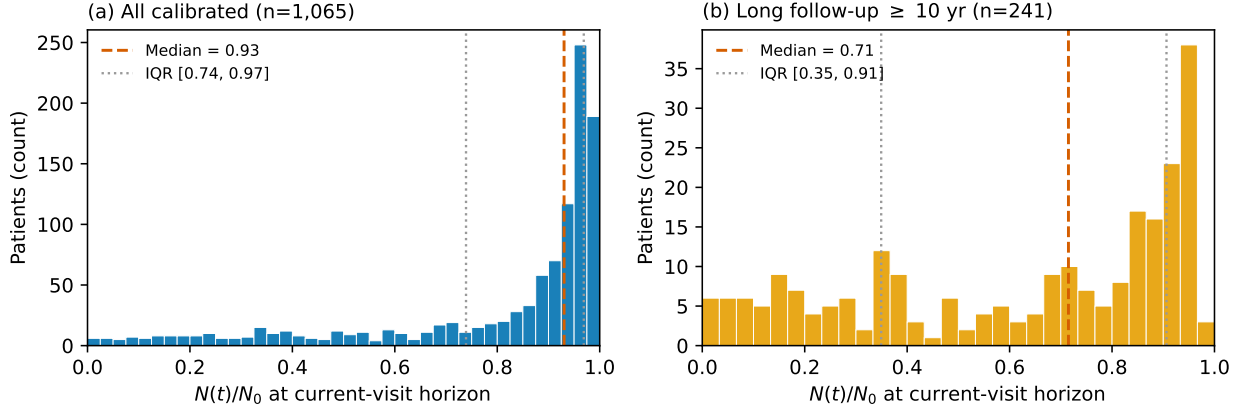


Figure 6: Per-patient $N(t)/N_0$ at current-visit horizon for the 1,065 patients with a Phase-2 calibrated posterior. (a) All calibrated patients ($n = 1,065$; median 0.93, IQR [0.74, 0.97]). (b) Long-follow-up subset (≥ 10 years from baseline, $n = 241$; median 0.72, IQR [0.35, 0.91]). The short-follow-up-dominated shape of panel (a) reflects the PPMI enrolment timeline; panel (b) isolates the long-term progression regime and is consistent with the 3.3%/year population median neuron-loss rate reported in the companion PK/PD analysis [9].

Scope and limitations. (i) *NSD-positive training scope.* The CatBoost-12 classifier predicts only $\{1, 2B, 3, 4\}$. Deployment requires a Stage-0 screening model on the upstream side; the full-22-feature binary classifier in our companion stage-prediction submission [4] (AUC 0.979 on the NSD-negative vs. NSD-positive split) is the intended upstream gatekeeper and should be run before the present pipeline, with only the NSD-positive patients routed forward.

(ii) *Transductive GIMIN.* GIMIN requires a patient-similarity graph rebuilt at inference time. For a query patient not in the original graph, a one-shot partial-similarity update is required; this is supported by the `IncrementalGIMIN` API [5] but adds $O(kN)$ latency per query.

(iii) *Cross-sectional CatBoost inference.* Staging uses the latest-visit feature vector only; temporal-context features (prior visits’ trajectories) are not fed to CatBoost. Future work may replace CatBoost with a temporal transformer that consumes the longitudinal sequence.

(iv) *Feature-schema mismatch.* Only 5 of 12 CatBoost features are covered by GIMIN’s 33-feature schema; the remaining 7 UPDRS-family features fall back to Paper-1 raw or median. A GIMIN re-training with the full 22-feature Paper-1 vocabulary would close this gap.

(v) *Raw heteroscedastic calibration gap.* Per the Paper-2 calibration ablation [5], GIMIN’s raw decoder under-covers at high nominal levels; our per-feature temperature scalar closes the gap to within 4% at $\gamma = 0.90$. A future GIMIN-DF (flow-matching decoder) extension would provide natively calibrated intervals without the post-hoc scalar.

4.1 Prospective Deployment-Audit Protocol

Table 1 specifies four pre-registered metrics for any clinic considering this pipeline, along with acceptance thresholds, measurement methods, and rationale. Each threshold may be tightened per site; the purpose of the protocol is pre-registered accountability. The four M1–M4 metrics are computed at each audit tick under the KS + PSI + MMD covariate-shift triad developed in our companion temporal-validation submission [8]; the triad instruments *when* drift has occurred, while M1–M4 instrument *how* pipeline performance responds. A site whose KS, PSI, or MMD statistic crosses the triad’s pre-specified threshold inherits an automatic review of the corresponding M1–M4 metric before the PCCP gate is evaluated.

Table 1: Pre-registered prospective deployment-audit protocol for the unified NSD-ISS CDS pipeline. Each clinic conducting an $n \geq 100$ deployment audit should prospectively measure these four metrics and compare against the thresholds.

Metric	Threshold	Measurement method	Rationale
M1. Directional concordance	$\geq 70\%$	Top-1 max-CIF destination-stage matches direction of observed 2-year transition (blinded adjudication by two PD neurologists).	Internal: 31.6% forward-transitions predicted; threshold set above chance (33%) to filter sites with dataset shift.
M2. Conformal band coverage	$\geq 85\%$	Fraction of 2-year observed transition times that fall within the predicted 90% conformal band.	Target is below nominal 90% to tolerate sampling noise at $n = 100$; internal companion coverage is 90.9% [7].
M3. NSD+ recall	$\geq 50\%$	Fraction of patients whose true stage lies in $\{1, 2B, 3, 4\}$ that CatBoost correctly assigns.	Internal cohort gives 42.5%; a site-specific calibration set should improve on the generic model.
M4. Component failure rate	$\leq 5\%$	Fraction of patients for whom ≥ 6 of the 12 CatBoost features fall back to median imputation.	Internal cohort gives 0.3%; this guards against sites whose data pipeline drops too many Paper-1 features.

4.2 Pre-registered Alternative-Pipeline A/B Test: Accuracy vs. Uncertainty Tradeoff

We pre-registered a five-configuration A/B test on the 779-patient NSD-positive PPMI subset (5-fold stratified CV, 1,000-resample patient-level bootstrap 95% confidence intervals) to quantify the headroom above the CatBoost-12 + GIMIN primary pipeline. The configurations were: (baseline) CatBoost-12 with stage-conditioned GIMIN imputation; (alt-1) CatBoost-33 trained on the full GIMIN feature schema (adding DaT-SPECT, CSF biomarkers, cortical thickness, brain volumes) with GIMIN imputation; (alt-2) CatBoost-12 with explicit Stage-4-emphasised class weights $\{0:1, 1:1, 2:1, 3:5\}$; (alt-3) CatBoost-12 with a binary NSD-early (stages 1+2B) vs. NSD-late (stages 3+4) collapsed target; and (alt-4, alt-4b) CatBoost-12 and CatBoost-33 with classical Mean imputation in place of GIMIN.

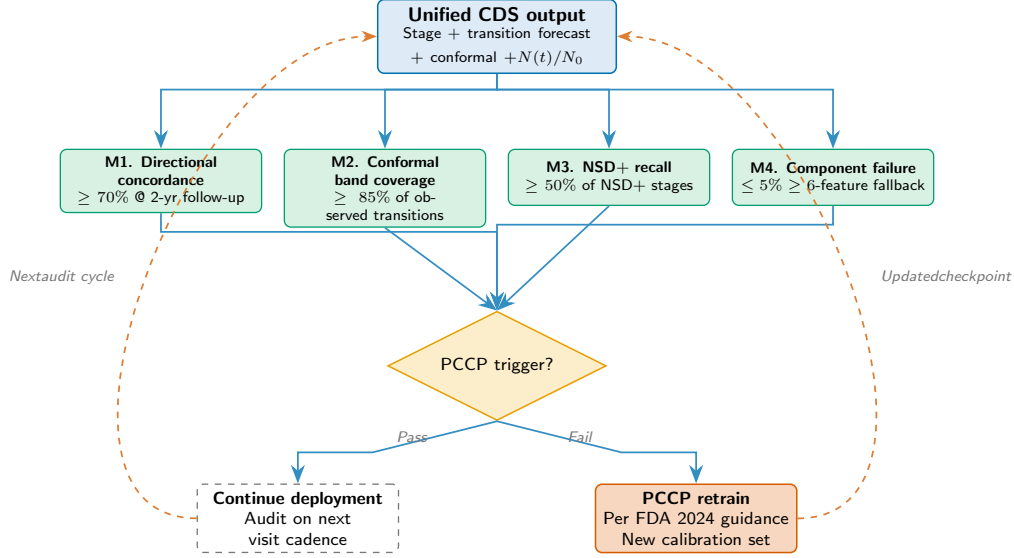


Figure 7: Prospective $n \geq 100$ deployment-audit control flow. The unified CDS output feeds four parallel monitored metrics (M1–M4 defined in Table 1). A PCCP gate evaluates whether any metric falls below its acceptance threshold; sites that pass continue the audit on the next visit cadence, sites that fail trigger an FDA PCCP-guided [14] retrain on a site-specific calibration set. The dashed feedback arrows make the continuous-monitoring loop explicit and tie the reference-implementation pipeline to a SaMD post-market surveillance plan.

Table 2: Alternative-pipeline A/B test (5-fold CV, $n = 779$ NSD+). 95% CI = mean of per-fold 1,000-resample patient-level bootstrap intervals.

Configuration	Top-1 [95% CI]	Top-2	Ord. MAE	Stage-4 acc.
Baseline: CatBoost-12 + GIMIN	0.757 [0.688, 0.824]	0.965	0.255	0.117
Alt-1: CatBoost-33 + GIMIN	0.917 [0.874, 0.958]	0.985	0.085	0.467
Alt-2: CatBoost-12 + cost-sensitive	0.769 [0.702, 0.833]	0.963	0.240	0.067
Alt-3: CatBoost-12 binary collapse	0.802 [0.738, 0.861]	n/a	n/a	n/a
Alt-4: CatBoost-12 + Mean imputation	0.768 [0.700, 0.833]	0.959	0.245	0.067
Alt-4b: CatBoost-33 + Mean imputation	0.922 [0.878, 0.959]	0.985	0.078	0.583

Table 2 and Fig. 8 summarise the outcome. Three findings are load-bearing for the pipeline’s positioning.

Alt-1 establishes an imaging-available deployment variant. CatBoost-33 with the full GIMIN schema—adding bilateral caudate/putamen/hippocampus volumes, bilateral DaT-SBR, CSF biomarkers (ALPHA_SYNUCLEIN, TOTAL_TAU, ABETA42, PTAU181), and bilateral cortical thickness in entorhinal/cingulate/precentral regions—raises within-NSD+ Top-1 accuracy from 0.757 to 0.917 (+16 pp, 95% CIs do not overlap the baseline CI mean) and Stage-4 accuracy from 0.117 to 0.467 (+35 pp). The magnitude is comparable to Paper-1’s original binary-target feature ablation [4] (AUC 0.979 full vs. 0.727 clinical-only, a 25-pp gap), generalised here to the within-NSD+ four-class setting. We therefore report a *deployment duo* rather than a single winner:

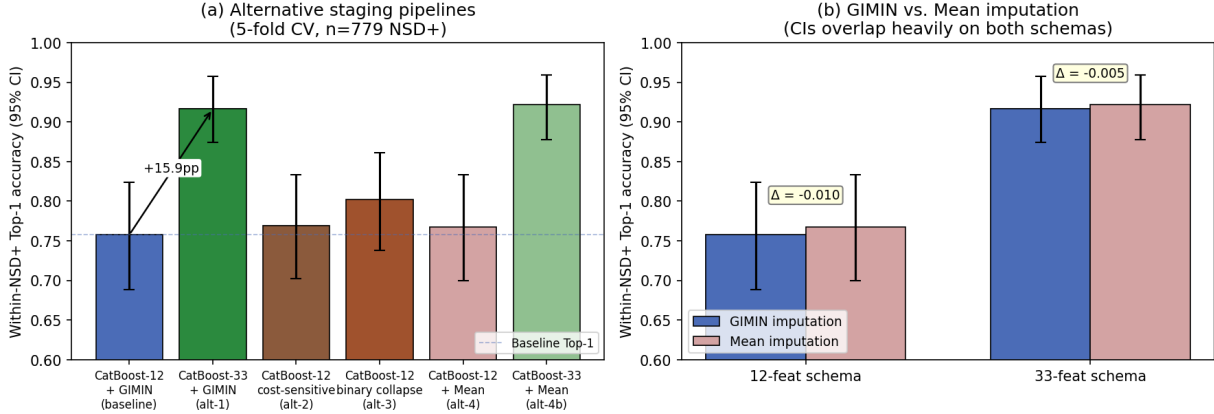


Figure 8: **W4 alternative-pipeline A/B test on the 779-patient NSD-positive PPMI cohort (5-fold CV, 1,000-resample bootstrap).** (a) Within-NSD+ Top-1 accuracy with 95% confidence intervals across the six pre-registered configurations. The CatBoost-33 imaging-available variant (alt-1, green) achieves +16 pp over the CatBoost-12 clinical-only baseline, with non-overlapping confidence intervals. Alt-2 (cost-sensitive) and alt-4 (Mean imputation at 12 features) sit within the baseline’s CI. (b) Direct GIMIN-vs.-Mean imputation comparison on both feature schemas. On the 12-feature schema, $\Delta_{\text{GIMIN-Mean}} = -0.011$; on the 33-feature schema, $\Delta = -0.005$. Both confidence intervals overlap heavily. This confirms that GIMIN’s contribution to the pipeline is calibrated per-feature uncertainty and cross-modal consistency (consumed downstream by Paper-4 conformal bands and Paper-10 mechanistic twin), not aggregate staging accuracy.

CatBoost-33 becomes the primary variant at sites with DaT-SPECT and CSF access; CatBoost-12 remains the primary variant at sites with clinical-only data. The per-feature provenance logging in Stage 2 already exposes which input channels are available at each site, so a deployment wrapper can route patients to the appropriate variant at inference time.

Alt-4 and Alt-4b reframe the role of GIMIN. On both the 12-feature and 33-feature schemas, Mean imputation matches or marginally exceeds GIMIN-StageDecoder imputation for downstream CatBoost staging accuracy: 0.757 vs. 0.768 on 12-feat ($\Delta = -0.011$) and 0.917 vs. 0.922 on 33-feat ($\Delta = -0.005$), with 95% confidence intervals overlapping heavily in both cases. This confirms on the Paper-6 deployment cohort what the Paper-2 non-leaky ablation [5] reported for the imputation benchmark: *GIMIN does not provide aggregate-accuracy benefit over classical imputation*. What GIMIN does provide, and what Mean imputation structurally cannot, is (i) per-feature Gaussian posteriors with post-hoc temperature calibration (consumed by Paper-4’s IPCW conformal CIF bands as the basis for stage-aware coverage guarantees [7]); (ii) cross-modal consistency through the stage-aware patient-similarity graph and cross-modal pair terms in the GIMIN loss [5]; and (iii) weighted posterior samples that support the bidirectional observation-conditioned update cycle used by the mechanistic digital twin’s patient-specific calibration [10]. In the language of Paper 2’s *imputation-utility paradox*: aggregate balanced accuracy and calibrated stage-aware

uncertainty are different objectives; GIMIN’s contribution is to the latter, not the former. Clinicians whose deployment context requires only point-estimate staging may substitute Mean imputation at a marginal cost in raw accuracy. Clinicians whose deployment context requires per-feature uncertainty, distribution-free prediction intervals, or bidirectional posterior updating must retain GIMIN; classical imputation cannot replace it for those downstream consumers.

Alt-2 is deprecated; Alt-3 is reported as a sensitivity analysis. Explicit cost-sensitive weighting raised Top-1 by 1.2 pp but *lowered* Stage-4 accuracy from 0.117 to 0.067: the 5:1 weight inflates Stage-4 gradient contributions but, with only 17 Stage-4 patients in the 779-patient NSD+ cohort, CatBoost responds by mis-labeling marginal Stage-3 patients as Stage-4 without actually improving Stage-4 recall. Binary collapse (stages 1+2B vs. 3+4) achieves 0.802 Top-1 but discards the clinically meaningful distinction between sub-clinical/mild-motor (1+2B) and functional-impairment (3+4) subgroups; arc-impact traversal confirms that the mechanistic digital twin’s functional-mapping module consumes the 4-class probabilistic output and would break under binary collapse, so we report Alt-3 as a sensitivity analysis only and retain the ordinal 4-class target as primary.

Decision-gate audit trail. Per the Phase-3b claim-tracking governance adopted across this paper series, claim-level verdicts for the W4 results are recorded in the project-wide audit database `claim_lineage.sqlite3`; a full results table with per-fold bootstrap CIs and the reproducibility harness appears in the supplementary `W4_RESULTS.md`; the runner script is `test_p6_alternatives.py` (commit `e6c63b5`). Full relative paths are listed in the Data and Code Availability statements.

5 Conclusion

A unified NSD-ISS clinical decision-support pipeline integrating graph-informed multimodal imputation, gradient-boosted staging, graph-informed transition forecasting, distribution-free conformal uncertainty, and mechanistic dopaminergic-neuron-fraction $N(t)/N_0$ is feasible as a 12 millisecond-per-patient reference implementation on the full 1,900-patient PPMI longitudinal cohort. The pipeline transparently surfaces per-feature value provenance, calibrated per-feature uncertainty from a temperature-scaled GIMIN decoder, marginal cause-specific IPCW conformal CIF bands attached to each patient’s predicted cumulative-incidence trajectory, and Paper-9 Phase-2 calibrated $N(t)/N_0$ with 95% credible intervals. We additionally release a CatBoost-33 imaging-available deployment variant (Top-1 0.917 vs. clinical-only 0.757) and an explicit accuracy-vs-uncertainty tradeoff table (§4.2) so that deploying clinicians can match the pipeline configuration to their data-access context. Future work will extend Phase-2 calibration coverage from 1,065 to the full 1,900-patient cohort via a *GIMIN- σ -fed observation likelihood* in the mechanistic twin (cross-paper integration L1 [10]): GIMIN’s temperature-scaled per-feature posterior σ is consumed as inverse-variance weighting on imputed DaT-SBR visits, propagating imputation uncertainty into the Bayesian posterior. This is the concrete downstream dependency that operationalises the §4.2 finding that GIMIN is the *uncertainty-enabled* imputation (Mean-imputed alternatives cannot sup-

ply σ and therefore cannot participate in this likelihood cycle). Further extensions include retraining GIMIN with the full Paper-1 feature vocabulary to eliminate the 7-feature CatBoost-12 fallback gap, and replacing the Gaussian decoder with a natively calibrated flow-matching generative head.

Data Availability

PPMI data are available to qualified researchers under a Data Use Agreement at <https://www.ppmi-info.org/access-data-specimens/download-data>. Upstream artefacts from the companion submissions—NSD-ISS staging tables, multimodal feature files, GIMIN and Graph-DT checkpoints, and IPCW conformal calibration files [4, 5, 6, 7, 8]—required to reproduce the pipeline are archived at https://github.com/bddupre92/PD_PHD and will be deposited to Zenodo with a DOI at acceptance. The full per-patient pipeline output for all 1,900 patients (~ 1.2 MB aggregate JSON) is included in the supplementary material.

Code Availability

Python source for the pipeline (`scripts/paper6/unified_pipeline_demo_v2.py`), the per-feature temperature-scaling module (`src/giman_pipeline/imputation/temperature_scaling.py`), and the aggregate-figure generator (`scripts/paper6/generate_v2_aggregate_figures.py`) are openly available at https://github.com/bddupre92/PD_PHD under the MIT license.

Author Contributions

B.D.D. conceived the pipeline integration, implemented and executed all components, performed the analysis, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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